

Dongho Kim

✉ Email 🌐 Website 🔗 LinkedIn

Research Interests

Reliable AI, Sociotechnical Evaluation of AI Systems, Human–AI Collaboration, Computational Social Science

Education

Northwestern University

M.S. in Statistics and Data Science

University of California, San Diego

Exchange Student, Business Economics and Data Science

Dongguk University

B.B.A. in Business Administration; B.S. in Data Science Software

Evanston, IL

Sep 2025 – Expected Jun 2027

San Diego, CA

Sep 2022 – Jun 2023

Seoul, South Korea

Mar 2019 – Feb 2025

Research Experience

Graduate Research Intern

Sep 2025 – Present

MAGICS Lab, Northwestern University; Advisor: Prof. Han Liu

Evanston, IL

- Co-developed SPINE (co-first author), an agentic harness that walks robotics novices through diagnosing and repairing real robots, with step-by-step checks until live validation passes.
- Designed the fault scenarios and the controlled comparison against human operators using Claude Code; novices with SPINE resolved every implanted fault across 12 scenarios and cut mean recovery time by 30–38%.
- Currently studying AI agents that automate research workflows (autoresearch) and agentic AI for science.

Research Collaborator

Oct 2025 – Present

NSF–Simons SkAI Institute, Northwestern University

Evanston, IL

- StarEmbed: benchmarked zero-shot embeddings from time-series foundation models (TSFMs) on ~40,000 expert-labeled variable-star light curves for classification and clustering.
- Pretrained a Transformer-based TSFM on irregularity-augmented data and explored timestamp-based rotary position embeddings and Mamba to address its limits on irregular sampling.

Research Assistant

Dec 2023 – Aug 2024

Intelligence and Optimization in Networks Lab; Advisor: Prof. Hyeryung Jang

Seoul, South Korea

- Implemented a hybrid RNN–FinBERT pipeline for stock-return prediction across 88 firms, combining price histories with text-derived sentiment features.

Research Projects

Sci2Pol-Agent: Agentic Workflows for Science-to-Policy Writing

Mar 2026 – Jun 2026

IEMS 447: Emerging Topics in Computational Social Science, Prof. Dashun Wang

Evanston, IL

- Prototyped training-free, multi-role agent workflows that route science-to-policy writing through evidence extraction, claim verification, and revision, and compared them with direct prompting on Sci2Pol-Bench.
- Follow-up (ongoing): measuring whether LLM-based reviewers catch unsupported or overstated claims in generated policy briefs.

Information Contagion and Market Connectivity

Sep 2025 – Dec 2025

COMP SCI 496: Social Network Analysis, Prof. Noshir Contractor

Evanston, IL

- Constructed a news co-mention network of 117 NASDAQ technology firms (2019–2020) and estimated volatility contagion with panel regressions and a stochastic actor-oriented model (RSiena).

Publications and Preprints (* equal contribution)

- Ham, M.*, **Kim, D.***, Lee, C.*, Kim, M. J., Zhang, Y., Wang, J., and Liu, H. *SPINE: Bridging the Cyber-Physical Gap with Agentic AI*. arXiv:2607.13049, 2026. Under review. Project page.
- Li, W.*, Chen, H.-Y.*, Rehemtulla, N.*, Shah, V., **Kim, D.**, Wu, D., Lin, Q., Miller, A., and Liu, H. *StarEmbed: Benchmarking Time Series Foundation Models on Astronomical Observations of Variable Stars*. ICML 2026 (poster); oral presentation and Outstanding Paper Award, AI4Physics Workshop at ICML 2026; arXiv:2510.06200.

Presentations

Frontier Agents for Variable-Star Analysis and Classification

Poster presentation, Open SkAI 2026; SkAI Institute

Sep 2026

Chicago, IL

Technical Skills

Programming: Python, PyTorch, R, SQL, Git, LaTeX

Machine Learning: Model pretraining and fine-tuning, LLM agent development, benchmark design and ablations

Statistics and Data: Regression, hypothesis testing, feature engineering, experimental design, network analysis

Professional and Teaching Experience

Teaching Assistant

Department of Business Administration, Dongguk University; Prof. Sanghyeon Sung

Sep 2023 – Jun 2024

Seoul, South Korea

- Courses: Organizational Behavior, Human Resource Management, and Diversity Management.

Operations Assistant

M&E Parts

Nov 2021 – Mar 2022

Seoul, South Korea

- Built an inventory analytics system to monitor stock levels, identify excess inventory, and support supply planning.

Sergeant, Military Service

Republic of Korea Ministry of National Defense

Jan 2020 – Sep 2021

South Korea

- Led a team managing mission-critical communication equipment and standardized operational-readiness reporting.

Awards and Scholarships

- Outstanding Paper Award, AI4Physics Workshop at ICML 2026 (oral presentation), 2026 – *StarEmbed*
- Excellence Award, Seoul City AI Video Content Competition, 2024
- Top 3%, LG AI Research B2B Sales Opportunity Prediction Competition, 2024
- Exchange Program Scholarship, Dongguk University, 2023
- Academic Excellence Scholarship, Dongguk University, 2022

Service and Languages

- International Student Mentor, International Buddy Program, Dongguk University, 2019
- Korean: native; English: proficient (GRE Q 170, V 158; IELTS 7.0); Chinese: intermediate (HSK 4)